EXPERIMENT-02

Aim: Convert the ER diagram into a relational schema. Write complete MySQL, CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

Relational Schema :

Customer(
CustomerID(PK),
Name,
Email UNIQUE,
Phone No.,
DOB
)

Address (
AddressID (PK),
CustomerID (FK),
Pincode,
State,
City
)

Seller (
SellerID (PK),
Seller Name,
Phone No. UNIQUE
)

Category (
CategoryID (PK),
CategoryName UNIQUE,
Description
)

Product (
ProductID (PK),
CategoryID (FK),
Price,
Image
)

Electronic (
ProductID (PK),
FK
)

Fashion (
ProductID (PK),
FK
)

Books (
ProductID (PK),
FK,
SellerID (FK)
)

Orders (
OrderID (PK)
CustomerID (FK),
OrderDate,
OrderStatus
)

Order_Item (
OrderID (PK/FK)
Item-seq-no (PK),
ProductID (FK)
Quantity,
Unit_Price
)

Payment (
PaymentID (PK),
OrderID (FK) UNIQUE,
PaymentDate,
Amount,
PaymentMethod,
Status
)

Delivery (
deliveryID (PK),
OrderID (FK) UNIQUE
CourierName,
Status,
DeliveryDate
)

Source Code -

```
CREATE DATABASE ECOMMERCE;
USE ECOMMERCE;
CREATE TABLE Customer(
        CustomerID INT Primary Key,
        Name Varchar(50) NOT NULL,
        Email Varchar(100) NOT NULL UNIQUE,
        Phone No. Varchar(15) NOT NULL,
        DOB DATE NOT NULL
);
CREATE TABLE Address(
        AddressID INT Primary Key,
        CustomerID INT NOT NULL,
        Pincode Varchar(10) NOT NULL,
        State Varchar(50) NOT NULL,
        City Varchar(50) NOT NULL,
        (FOREIGN KEY(CustomerID)
         references Customer(CustomerID)
         ON DELETE CASADE)
    );
CREATE TABLE Category(
        Category INT Primary Key,
        Category Name Varchar(100) NOT NULL,
        Description Varchar(255)
);
CREATE TABLE Seller(
        SellerID INT Sello Primary Key,
        Seller Name Varchar(100) NOT NULL,
        PhoneNo Vardhan(15) NOT NULL UNIQUE
);
```

```
CREATE TABLE Product(
        ProductID INT Primary Key,
        CategoryID INT NOT NULL,
        Price Decimal (10, 2) NOT NULL,
        Image Varchar (255),
       (Foreign Key (CategoryID) references
        Category (CategoryID) ON DELETE
        CASCADE)
);
CREATE TABLE Orders(
        OrderID INT Primary Key,
        CustomerID NOT NULL,
        OrderDate DATE NOT NULL,
        OrderStatus Varchar(30) NOTNULL,
        (Foreign Key (CustomerID) references
        Customer (CustomerID) on DELETE
        CASCADE)
);
CREATE TABLE OrderItem(
        OrderID INT NOTNULL,
        Item-seq-no INT NOT NULL,
        ProductID INT NOT NULL,
        Quantity INT NOT NULL,
        Unit-Price DECIMAL(10,2) NOT NULL,
        Primary Key (OrderID, Item-seq-no),
       (Foreign Key (orderID) references
        Orders (OrderID) ON DELETE
        CASCADE)
       (Foreign Key (ProductID) references
        Product (ProductID) ON DELETE
        CASCADE)
);
```

Create TABLE Delivery (
DeliveryID INT Primary Key,
OrderID INT NOT NULL UNIQUE,
CourierName Varchar(100) NOT NULL,
Status Varchar(30) NOT NULL,
Delivery Date DATE,
Foreign Key (OrderID) references
Orders(OrderID) ON DELETE
CASCADE)
);

Inserting Data -

Insert into Customer(Name, Email, PhoneNo),
DOB) VALUES
("Shailendra", "Shailendra@gmail.com", "9856717891",
"2001-05-15"),
("Ram", "Ram@gmail.com", "9331567891", "2006-04-20");
Insert into Category (CategoryName, Description)
Values ("Electronics", "Electronic Product"),
("Books", "Book");

Insert into Seller(SellerName, PhoneNo)
Values ("Tech Store", "8000016172"),
("Book Store", "7000175412 3");

Insert into Product(CategoryID, Price, Image)
Values(1, 499, "PC.jpg"), (2, 599, "Pant.jpg"),
(3, 699, "book.jpg");

Insert into Electronics Values (1);
Insert into Fashion Values (2);
Insert into Books Values (3, 2);

Insert into Address (CustomerID, Pincode, State, City) Values
(1, "208001", "Uttar Pradesh", "Kanpur"),
(2, "110001", "Delhi", "Delhi");

Insert into Orders (CustomerID, OrderDate, OrderStatus) Values
(1, "2026-08-19", "Placed"),
(2, "2026-08-19", "Shipped");

Insert into OrderItem Values
(1, 1, 1, 1, 499), (2, 1, 2, 2, 599);

Insert into Payment (OrderID, PaymentDate, Amount, PaymentMethod, Status) Values
(1, "2026-08-19", 50000, "UPI", "Success"),
(2, "2026-08-19", 19981, "Card", "Success");

Insert into Delivery (OrderID, CourierName, Status, DeliveryDate) Values
(1, "Delhivery", "Delivered", "2026-08-20"),
(2, "DTDC", "Pending", NULL);

```
Select * from Customer;
Select * from Product;
Select * from Orders;
Select * from OrderItem;
```

```
ON DELETE SET NULL
DELETE FROM Seller where SellerID=2;
Select * From Books;
```

```
ON DELETE CASCADE
DELETE From Customer where
  CustomerID=2;
Select * from orders where CustomerID=2;
```

OUTPUT :

CustomerID	Name	Email	Phone No	DOB
1	Shailendra	Shailen[illegible]gmail.com	9856717842	2001-03-15
2	Ram	Ram@gmail.com	9331567891	2006-04-20

ProductID	CategoryID	Price	Image
1	1	499	PC.Png
2	2	599	Pant.Png
3	3	699	Book.Png

OrderID	CustomerID	OrderDate	OrderStatus
1	1	2026-08-14	Placed
2	2	2026-08-19	Shipped

OrderID	ItemSeqNo	ProductID	Quantity	UnitPrice
1	1	1	1	499.00
2	1	2	2	599.00

ProductID	SellerID
3	NULL

EMPTY SET

